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Editor
The Journal of Chemical Physics

Dear Editor,

Please find enclosed the manuscript "Algorithmic Construction and Exploitation of Topology-Constrained Refinement Models for Semiexperimental Equilibrium Structures," submitted for consideration as an Article in *The Journal of Chemical Physics*.

The manuscript introduces MORPHEUS, a framework that turns the formulation of semiexperimental refinement models into a reproducible topology-, symmetry-, and rank-audited procedure. Rather than assuming that the internal-coordinate model has already been selected by expert intervention, MORPHEUS uses SMITH-generated SONIC coordinate contracts, applies symmetry and homogeneous rank reduction, projects constraints and predicates consistently, and propagates uncertainties back to chemically interpretable structural parameters. When an internal-coordinate correction must be realized, MORPHEUS uses a rank-controlled minimum-norm or nonlinear internal-to-Cartesian realization, without introducing a second Wilson-B inversion algorithm.

The work is, in my view, well aligned with the scope of *The Journal of Chemical Physics* because it addresses a general problem at the interface of rotational spectroscopy, molecular structure determination, and quantum chemistry: how to combine high-resolution experimental data, computed vibrational and electronic corrections, topology, symmetry, and external reference information in a controlled statistical model. The validation spans acyclic, cyclic, planar, fused, bridged, and flexible molecules. A large-molecule testosterone case study further shows that accurate quantum chemical equilibrium structures, accelerated vibrational corrections, and algorithmically generated topology-constrained refinement models can be chained for systems containing about 50 atoms, where conventional isotope-rich semiexperimental analyses are not currently practical.

The manuscript also emphasizes reproducibility. The reported models are not defined by molecule-specific Z-matrix choices, but by explicit primitive relations, parameter classes, predicates, and audit trails. The submitted Supporting Information documents the benchmark inputs, diagnostics, and the large-molecule class/predicate variants used in the paper.

The coordinate-generation layer used here is the subject of the companion SMITH/SONIC methodological preprint, publicly available as arXiv:2607.16550.

This manuscript is original, has not been published previously, and is not under consideration elsewhere. I confirm that there are no conflicts of interest to declare.

Sincerely,

Vincenzo Barone